



Learning Shared Economic Representations through the Application of Multi-Task Gradient Boosting

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Abstract

Machine learning methods are increasingly used in economics, yet most applications rely on single-task approaches that model outcomes, markets, or regions in isolation. In this study, we adopt a multi-task learning (MTL) framework for economic and tabular data to analyze interdependencies across related economic units. Using benchmark housing datasets representing different economic environments, we define each task as a regional or categorical subgroup. The MTL approach exploits shared representations while allowing for task-specific effects, leading to improved predictive performance and revealing latent relationships across subgroups. An empirical evaluation across three datasets demonstrates the suitability of MTL for structured economic data. Overall, the results suggest that MTL provides a transparent and flexible framework for jointly modeling related entities such as regions, industries, or firms.

Keywords Multi-task learning · Gradient boosting · Computational economics · Cross-regional modeling · Structured economic data

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1 Introduction

Machine learning (ML) models have become increasingly prevalent in empirical economics (Hara, 2025; Hopp, 2024; Hu et al., 2022; Li et al., 2022; Shahin et al., 2025), enabling the modeling of high-dimensional and non-linear relationships among economic variables (Ma et al., 2020; Scheidegger & Bilonis, 2019; Tang & Mei, 2026). However, most econometric applications have focused on single-task learning paradigms (Athey & Imbens, 2019; Mullainathan & Spiess, 2017; Varian, 2014); in which each predictive problem is treated independently. As a result, econometricians typically need to train separate models for each task (e.g., outcome, region, or segment) (Gu et al., 2021; Hopp, 2024; Ma et al., 2020), disregarding potential interdependencies among related economic features. This independence assumption can lead to inefficiencies in learning, especially in cases where features share latent structures (Caruana, 1997; Zhang & Yang, 2022). Moreover, training separate models for each task can result in redundant computations and increased complexity, hindering interpretability and generalization (Scheidegger & Bilonis, 2019).

Multi-task learning (MTL) provides a natural framework to overcome these limitations by allowing related predictive problems to be learned jointly rather than independently (Caruana, 1997; Zhang & Yang, 2022). Through the sharing of representations across tasks, MTL can exploit common information while preserving task-specific components, thereby improving generalization (Caruana, 1997), particularly in domains characterized by structured and heterogeneous data (Zhang & Yang, 2022). Such properties make MTL especially suitable for economic applications, where outcomes across regions, markets, or firms often depend on shared underlying mechanisms and correlated explanatory variables. To the best of our knowledge, MTL has not yet been systematically applied within economics, especially in structured or tabular micro-level economic entities such as regions, markets, or firms. While MTL has shown promising results in related domains such as energy forecasting, manufacturing prediction, and economic indicator analysis (Mehta & Shao, 2021; Song et al., 2024; Ying et al., 2025), its potential for economic data remains largely unexplored.

Various MTL approaches have been proposed in the ML literature, including neural network-based methods (Caruana, 1997; Zhang et al., 2014), Bayesian models (Bakker & Heskes, 2003; Yu et al., 2005), and ensemble methods (Chapelle et al., 2011; Emami et al., 2023, 2026). Among these, ensemble-based approaches—specifically Gradient Boosting (GB) (Friedman, 2001)—have demonstrated strong performance on tabular data (Bentéjac et al., 2021; Schwartz-Ziv & Armon, 2022), making them especially well-suited for structured economic datasets and related applications. The present study applies the Multi-Task Gradient Boosting (MTGB) framework introduced by Emami et al. (2023), which integrates shared and task-specific decision tree ensembles. While MTGB has demonstrated empirical success in general machine learning applications (Emami et al., 2023), it has not yet been explored within the context of data-driven economic modeling. This work extends its use to structured economic datasets to evaluate the ability of MTGB to capture both shared and task-specific patterns across related economic subgroups, thereby assessing its potential within the field of computational economics.

Our contributions to the field of computational economics are threefold:

- (i) **Empirical application of multi-task learning to economics.** The recently proposed MTGB methodology (Emami et al., 2023) is applied to structured economic datasets, demonstrating how an established MTL approach can be used to model related economic categories jointly rather than independently.
- (ii) **Systematic evaluation on tabular economic data.** Using benchmark housing datasets, we apply MTGB to a multi-task setting where each task corresponds to a distinct economic or structural subgroup. MTGB is then compared with state-of-the-art single-task boosting algorithms to assess its empirical performance.
- (iii) **Providing interpretability and economic insights.** The shared and task-specific components learned by MTGB reveal common and subgroup-specific determinants of housing prices, offering a transparent, interpretable, and policy-relevant framework for economic modeling.

The remainder of this study is organized as follows. Section 2 reviews related work on ensemble boosting methods, MTL, and applications of MTL in economics, highlighting connections and differences with the present study. Section 3 presents the methodological framework by reviewing gradient boosting-based models, including the MTGB approach proposed by Emami et al. (2023), as well as the single-task boosting methods used for comparison. Section 4 presents the experimental design, including datasets, baseline models, evaluation metrics, and implementation details, and reports the main empirical results. Section 5 provides an *Economic Interpretation and Insights*, where the task-level results are examined to extract economically meaningful patterns and to interpret the shared and task-specific learning components within the context of structural heterogeneity across submarkets. Finally, Section 6 concludes the paper and outlines directions for future research on multi-task ensemble learning in economics.

2 Related Work

Gradient Boosting (GB) is an ensemble learning method that constructs a strong predictor by sequentially combining base learners (i.e., decision tree regressors) (Friedman, 2001). Its high predictive performance and interpretability have led to the development of several popular variants, including XGBoost (Chen & Guestrin, 2016), LightGBM (Ke et al., 2017), and CatBoost (Prokhorenkova et al., 2018). XGBoost introduced regularization techniques to enhance generalization and prevent overfitting, LightGBM improved computational efficiency through histogram-based splitting and leaf-wise tree growth, and CatBoost addressed the challenges of categorical feature encoding and reduced prediction bias using ordered boosting.

Beyond single-output prediction, GB has been extended to handle multi-output problems in both classification and regression settings (Emami & Martínez-Muñoz, 2025; Ponomareva et al., 2017; Zhang & Jung, 2021), enabling base learners to produce vector-valued outputs and employing specialized loss functions to capture dependencies among multiple outputs, thereby improving predictive performance in settings with correlated outputs.

To address the challenge of learning from related tasks more effectively, MTL has emerged as a powerful paradigm that exploits shared representations. One of the first studies on MTL grouped related tasks into clusters (Thrun & O'Sullivan, 1996). They improved learning by using shared patterns within each cluster. Another study on MTL was conducted by Caruana (1997). The author showed that related tasks could improve performance by sharing common information within a single model. In this framework, a shared structure learns features across tasks, while each task maintains its own specific output. A low-rank approach to MTL was proposed by Ando et al. (2005), which assumes that the parameters for all tasks share a common low-dimensional structure. Later, a study proposed decomposing the model parameters for each task into shared components and task-specific differences, enabling multiple related tasks to be learned simultaneously (Evgeniou & Pontil, 2004).

Building on these MTL approaches, recent works have adapted GB to learn multiple related and unrelated tasks simultaneously (Chapelle et al., 2011; Emami et al., 2023, 2026). The first study to introduce a multi-task boosting approach was proposed by Chapelle et al. (2011). They built a system with one shared model for all tasks and separate models for each single-task. During training, the algorithm adds new models to whichever component improves performance the most. Conversely, a different approach known as MTGB has been proposed (Emami et al., 2023). It divides the learning process into two steps: first, learning shared patterns across tasks; second, learning unique adjustments for each task. The final prediction for a task combines both shared and task-specific components. Lastly, a more robust version of MTGB was introduced, capable of handling outlier tasks (Emami et al., 2026). The method includes a mechanism that detects and reduces the influence of outlier tasks during training.

In the economic and financial domain, MTL has been applied to different prediction and decision-making problems. One early application of MTL in finance examined stock selection across Canadian markets, where predicting returns for multiple stocks was treated as a set of related tasks (Ghosn & Bengio, 1996). Using artificial neural networks trained directly on a profit-maximizing objective, the study compared full, partial, and no parameter sharing between tasks. Results showed that partial sharing improved generalization and portfolio returns. Another application of MTL in finance presents a linear model optimized for profit maximization through MTL to capture both inter-company relationships and temporal dependencies in financial markets (Bitvai & Cohn, 2015). In their framework, each company is modeled as a separate task, and a mean-regularized multi-task learning approach (Evgeniou & Pontil, 2004) is employed to jointly train all tasks, enabling the model to share information across related firms while preserving task-specific characteristics. The most recent application of MTL in finance was presented by Prabhu et al. (2021), who used an artificial neural network to simultaneously predict the closing prices of multiple correlated stocks.

Based on our review of the literature, this is the first study to apply a multi-task ensemble model in the field of economics and finance. Specifically, it employs the MTGB framework proposed by Emami et al. (2023), which is based on ensemble learning rather than neural networks and is specifically designed for tabular datasets. This distinguishes the current study from prior studies that have predominantly

implemented neural network-based MTL architectures. Moreover, whereas existing research has concentrated on financial market indicators such as stock prices and returns, this study focuses on microeconomic, cross-sector datasets encompassing firm and household-level variables.

3 Baseline Boosting Models

This section provides an overview of the methodological foundations of the models adopted in this study. We consider a supervised learning problem in which the objective is to learn a prediction function $F : \mathbb{R}^d \rightarrow \mathbb{R}$ that maps economic covariates $\mathbf{x}_i$ to an outcome variable y_i by minimizing an expected loss function. Given the observed dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i, t_i)\}_{i=1}^N$, the empirical risk minimization problem is,

$$\min_F \sum_{i=1}^N \Psi(y_i, F(\mathbf{x}_i)),$$

where $\Psi(\cdot, \cdot)$ is a differentiable loss function measuring the discrepancy between observed outcomes and model predictions, $\mathbf{x}_i \in \mathbb{R}^d$ is a vector of economic covariates, $y_i \in \mathbb{R}$ is the outcome variable of interest, and $t_i \in \{1, \dots, T\}$ indexes the task associated with observation i .

Within this framework, the Gradient Boosting (GB) model proposed by Friedman (2001) is first summarized in Subsection 3.1, as it forms the basis of all boosting algorithms considered in this study. The Multi-Task Gradient Boosting (MTGB) approach proposed by Emami et al. (2023) is discussed in Subsection 3.2, where it extends the conventional single-task GB framework to multi-task settings by jointly modeling shared and task-specific components. For completeness and to ensure a transparent comparative analysis, the guiding principles of the single-task boosting models used as baselines (XGBoost (XGB) (Chen & Guestrin, 2016), LightGBM (Ke et al., 2017), and CatBoost (Prokhorenkova et al., 2018) are reviewed in Subsections 3.3–3.5. Finally, Subsection 3.6 provides a systematic comparison of these models, highlighting their conceptual differences, modeling assumptions, and suitability for single-task and multi-task learning settings.

3.1 Gradient Boosting

Gradient Boosting (GB), introduced by Friedman (2001), is an ensemble learning method that constructs a predictive function by sequentially adding base learners (i.e., decision tree regressors) to correct the residual errors of the model built so far. GB constructs a predictive function by solving the empirical risk minimization problem,

$$F(\mathbf{x}) = \arg \min_{F(\mathbf{x})} \mathbb{E}_{(y, \mathbf{x})} [\Psi(y, F(\mathbf{x}))].$$

The learning process follows an iterative strategy in which the objective function is approximated by an additive combination of base learners (h_m),

$$F(\mathbf{x}) = \sum_{m=0}^M \rho_m h_m(\mathbf{x}),$$

where M is the total number of base learners and ρ is the step size. The model is learned by successively updating the function in a stagewise manner,

$$(h_m, \rho_m) = \arg \min_{(\rho_m, h_m)} \sum_{i=1}^N \Psi(y_i, F_{m-1}(\mathbf{x}_i) + \rho_m h_m(\mathbf{x}_i)),$$

starting from the constant model $F_0 = \arg \min_{\rho} \sum_{i=1}^N \Psi(y_i, \rho)$. At each iteration, a decision tree regressor is fitted to the pseudo-residuals obtained from the negative gradient of the loss function,

$$r_{i,m} = - \left[\frac{\partial \Psi(y_i, F(\mathbf{x}_i))}{\partial F(\mathbf{x}_i)} \right]_{F=F_{m-1}}, \quad i = 1, \dots, N.$$

Specifically, a decision tree regressor h_m is trained on $\{(x_i, r_{i,m})\}_{i=1}^N$. Subsequently, the optimal step size is determined by performing a Newton-Raphson line search,

$$\rho_m = \arg \min_{\rho_m} \sum_{i=1}^N \Psi(y_i, F_{m-1}(\mathbf{x}_i) + \rho h_m(\mathbf{x}_i)).$$

Finally, the model is updated by incorporating the contribution of the newly fitted base learner,

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \nu \rho_m h_m(\mathbf{x}).$$

A further hyperparameter $\nu \in (0, 1]$ functions as a learning rate, governing the contribution of the base learner (decision tree regressor) to the ensemble model.

GB is widely used in computational economics (Liu et al., 2019; Yoon, 2021). However, it is inherently a *single-task* method: a separate model must be trained for each economic subgroup or region, preventing the exploitation of shared structure across tasks.

3.2 MTGB

This subsection summarizes MTGB algorithm introduced by Emami et al. (2023), which extends GB (Friedman, 2001) to handle multiple related tasks that share an input space. The aim is to exploit inter-task information by combining shared and

task-specific learning components. The following description is adapted from Emami et al. (2023) for completeness.

Let T denote the total number of tasks. The objective is to estimate a set of functions $\{F_t\}_{t=1}^T$ such that,

$$F_t(x) = f(x) + g_t(x),$$

where f captures the common structure and g_t accounts for task-specific functions.

In the MTGB framework, the final predictor for each task is composed of a shared set of T_c base learners and a task-specific set of $T_s = T - T_c$ base learners per task. The resulting prediction function for task t is given by

$$F_{M,t}(x) = h_0(x) + \sum_{m=1}^{T_c} \nu h_m(x) + \sum_{m=1}^{T_s} \nu h_m(x).$$

During training, the shared base learners are fitted using all tasks simultaneously by minimizing the aggregated loss,

$$h_m^{(c)} = \arg \min_h \sum_{t=1}^T \sum_{i:t_i=t} \Psi(y_{i,t}, F_{(m-1)}(x_{i,t}) + h_m(x_{i,t})),$$

while the task-specific base learners are optimized independently for each task,

$$h_m^{(s)} = \arg \min_{h_m^{(s)}} \sum_{i:t_i=t} \Psi(y_i, F_m(x_i) + h_m(x_{i,t})).$$

3.3 XGBoost

XGBoost, proposed by Chen and Guestrin (2016), is a regularized extension of GB framework, and designed to enhance generalization performance and computational efficiency in high-dimensional tabular data. Similar to standard GB, XGBoost constructs an additive model in a stage-wise manner. At training iteration m , the prediction function is defined as,

$$F_m(x) = F_{m-1}(x) + \nu h_m(x),$$

where ν is the learning rate.

The key distinction of XGBoost lies in its regularized optimization objective. Instead of minimizing the empirical loss alone, XGBoost minimizes the following regularized objective function at iteration m ,

$$\mathcal{L}_{(m)} = \sum_{i=1}^N [\Psi(y_i, F_{m-1}(x_i) + h_m(x_i))] + \Omega(h_m),$$

where $\Omega(h_m)$ is a regularization term penalizing model complexity,

$$\Omega(h_m) = \gamma\mathcal{T} + \frac{\lambda}{2} \sum_{j=1}^{\mathcal{T}} \omega_j^2,$$

with $\mathcal{T}$ denoting the number of terminal nodes (leaves) in the tree (h_m), and ω_j representing the prediction score associated with leaf j . The parameters $\gamma \geq 0$ and $\lambda \geq 0$ control the complexity of the tree structure and the magnitude of the leaf weights, respectively.

The objective in XGBoost is optimized using a second-order Taylor approximation of the loss evaluated at the current prediction,

$$\mathcal{L}_{(m)} \approx \sum_{i=1}^N \left[\Psi(y_i, F_{m-1}(x_i)) + g_i h_m(x_i) + \frac{1}{2} H_i h_m(x_i)^2 \right] + \Omega(h_m),$$

where,

$$g_i = \left[\frac{\partial \Psi(y_i, F(x_i))}{\partial F(x_i)} \right]_{F=F_{m-1}}, \quad H_i = \left[\frac{\partial^2 \Psi(y_i, F(x_i))}{\partial F(x_i)^2} \right]_{F=F_{m-1}}.$$

are the first- and second-order derivatives of the loss function with respect to the prediction, respectively.

The strong performance of XGBoost on tabular and panel data has made it a standard benchmark in applied computational economics (Shahin et al., 2024; Yu et al., 2021).

3.4 LightGBM

LightGBM, proposed by Ke et al. (2017), is a computationally efficient implementation of XGBoost developed for large-scale datasets commonly encountered in applied economics (Çoban & Ulusay, 2025; Wang & Jiang, 2023). Like XGBoost, LightGBM constructs an additive model of base learners, and optimizes a regularized objective using a second-order Taylor approximation of the loss function.

LightGBM introduces two key features, *Gradient-based One-Side Sampling* (GOSS) and *Exclusive Feature Bundling* (EFB), which enable efficient training on large-scale datasets. GOSS accelerates boosting by selecting instances according to the magnitude of their gradients, preserving information-rich samples while down-sampling less informative ones. Recall that the training data are $\{(x_i, y_i)\}_{i=1}^N$, and at boosting iteration m , the current prediction is $F_{(m-1)}$, with first- and second-order gradients of the loss function defined as $g_{(i,m)}$ and $h_{(i,m)}$ (See Subsection 3.3).

GOSS sorts instances by $|g_{(i,m)}|$ and retains the top a fraction,

$$A_{(m)} = \text{Top}(\{|g_{(i,m)}|\}_{i=1}^N, a),$$

while sampling uniformly from the remaining instances $(A_{(m)})_c$,

$$B_{(m)} \subset (A_{(m)})_c, \quad |B_{(m)}| = b|(A_{(m)})_c|.$$

To compensate for the under-representation of small-gradient instances, a reweighting factor is introduced,

$$\text{fact} = \frac{1-a}{b},$$

and instance weights are assigned as,

$$\phi_i^{(m)} = \begin{cases} 1, & i \in A^{(m)}, \\ \text{fact}, & i \in B^{(m)}. \end{cases}$$

For any subset of instances S , the weighted sums of gradients and Hessians are,

$$\mathcal{G}(S) = \sum_{i \in S \cap (A_{(m)} \cup B_{(m)})} \phi_{(i,m)} g_{(i,m)}, \quad \mathcal{H}(S) = \sum_{i \in S \cap (A_{(m)} \cup B_{(m)})} \phi_{(i,m)} H_{(i,m)}.$$

Using these weighted sums, LightGBM preserves the standard second-order split criterion of gradient boosting.

Complementing this instance-level optimization, LightGBM also addresses high-dimensional and sparse feature spaces through *Exclusive Feature Bundling* (EFB). In many applications, features are mutually exclusive in the sense that they rarely take nonzero values. EFB exploits this sparsity by grouping such features into a smaller number of bundles, each treated as a single composite feature during tree construction. By assigning disjoint value ranges to features within the same bundle, LightGBM preserves the information required for split decisions while substantially reducing the effective feature dimension.

3.5 CatBoost

CatBoost, developed by Prokhorenkova et al. (2018), addresses two major challenges in gradient boosting: the treatment of categorical variables and the prediction shift caused by target leakage. It relies on an *ordering principle* that is applied consistently to both categorical feature encoding and the boosting procedure. As in standard GB methods, CatBoost constructs an additive predictive function composed of decision tree base learners.

Recall that $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ denotes the training dataset, and let π be a random permutation of the indices $\{1, \dots, N\}$. CatBoost differs from conventional boosting approaches in the way categorical features are processed. Instead of one-hot encod-

ing, it employs *ordered target statistics*, which compute category-level estimates using only observations that precede the current sample in the permutation.

Formally, for a categorical feature $\mathcal{C}$, the encoded value for observation i is defined as a smoothed empirical estimate,

$$\tilde{\mathcal{C}}_i = \frac{\sum_{j: \pi(j) < \pi(i)} 1\{\mathcal{C}_j = \mathcal{C}_i\} y_j + \mathcal{A} p}{\sum_{j: \pi(j) < \pi(i)} 1\{\mathcal{C}_j = \mathcal{C}_i\} + \mathcal{A}},$$

where p is a prior value and $\mathcal{A} > 0$ is a smoothing parameter. Hence, the encoding of observation i does not depend on its own target value.

In addition, CatBoost implements *ordered boosting* to eliminate prediction shift during model training. Let $\pi(1), \dots, \pi(N)$ denote a random permutation of the training indices, where $\pi(\mathbf{p})$ is the observation at position $\mathbf{p}$ in the permutation. The algorithm maintains a sequence of supporting (prefix) models $\{h_{\mathbf{p}}\}_{\mathbf{p}=0}^N$, where $h_{\mathbf{p}}$ is trained using only the first $\mathbf{p}$ observations in the permutation and is initialized as $h_0(x) = 0$.

For each position $\mathbf{p} = 1, \dots, N$, the residual for observation $\pi(\mathbf{p})$ is computed using the model trained on the preceding prefix,

$$r_{\pi(\mathbf{p})} = y_{\pi(\mathbf{p})} - h_{\mathbf{p}-1}(x_{\pi(\mathbf{p})}),$$

so that the prediction used to compute the residual is obtained from a model trained without access to the target value $y_{\pi(\mathbf{p})}$. A base learner is then fitted by minimizing the empirical loss over the corresponding prefix,

$$\Delta h_{\mathbf{p}} = \arg \min_h \sum_{q=1}^{\mathbf{p}} \Psi(r_{\pi(q)}, h(x_{\pi(q)})),$$

and the prefix model is updated accordingly,

$$h_{\mathbf{p}} = h_{\mathbf{p}-1} + \Delta h_{\mathbf{p}}.$$

CatBoost has been applied in various economic settings, including credit scoring and financial risk assessment (Fan, 2025; Liu et al., 2024). Its ability to handle high-cardinality categorical features while reducing prediction bias makes it a strong baseline for tabular economic data.

3.6 Boosting Framework Comparison

Table 1 contrasts the boosting-based learning frameworks analyzed in this study by focusing on their algorithmic design, optimization strategies, and ability to capture cross-task information. All methods build on the gradient boosting paradigm, tree base learners and minimization of a differentiable loss function. GB (Friedman,

Table 1 Comparative summary of boosting models. (✓: feature present; ×: feature absent)

Feature	GB	MTGB	XGBoost	LightGBM	CatBoost
Boosting framework	✓	✓	✓	✓	✓
Decision-tree base learners	✓	✓	✓	✓	✓
Additive stage-wise learning	✓	✓	✓	✓	✓
Differentiable loss required	✓	✓	✓	✓	✓
Second-order optimization	×	×	✓	✓	×
Explicit regularization term	×	×	✓	✓	✓
Single-task learning	✓	✓	✓	✓	✓
Multi-task learning	×	✓	×	×	×
Native categorical handling	×	×	×	×	✓
Target leakage prevention	×	×	×	×	✓
Ordered boosting	×	×	×	×	✓
Economic benchmark	✓	×	✓	✓	✓

2001) provides the conceptual baseline but is limited to single-task estimation, requiring separate models for each task and preventing any sharing of information across related economic settings. XGBoost (Chen & Guestrin, 2016) and LightGBM (Ke et al., 2017), extend this framework with second-order optimization, explicit regularization, and more efficient tree construction, with LightGBM enhancing scalability and CatBoost (Prokhorenkova et al., 2018) addressing categorical variables and target leakage.

In contrast, MTGB (Emami et al., 2023) framework introduces a shared component alongside task-specific deviations, enabling joint optimization across tasks while preserving heterogeneity. This makes MTGB the only approach in the comparison that simultaneously exploits shared representations and task-level specialization. While the single-task algorithms excel in predictive performance and computational efficiency, they remain unable to leverage cross-task dependencies.

4 Experiments and Results

This section presents the baseline state-of-the-art models and the model configurations described in Subsection 4.1. In Subsection 4.2, an overview of the micro-level public economic datasets used in the experiments is provided, along with their key characteristics. The evaluation metrics used to assess the performance of the studied models are also described in Subsection 4.3. Finally, the results are reported and discussed in Subsection 4.4.

4.1 Baseline Models

To evaluate the performance of MTGB (Emami et al., 2023), a comparative analysis was conducted across public datasets and state-of-the-art ensemble boosting models, including LightGBM (Ke et al., 2017), GB (Friedman, 2001), XGBoost (XGB) (Chen & Guestrin, 2016), and CatBoost (Ponomareva et al., 2017). These approaches are inherently single-task models; therefore, they were trained independently for each task, and the final mean results were reported. In contrast, MTGB was trained simultaneously on all tasks, allowing them to be learned in parallel.

For all experiments, the datasets were divided into training and test sets using an 80:20 split. The hyperparameter tuning was performed on the training set using a grid search combined with 5-fold cross-validation. The grid included `Learning Rate` values of [0.05, 0.1, 0.5, 1.0] and `Max Depth` values of [3, 5, 7, 10, 20] for all models. The number of base learners was fixed at 100 for all models, while for MTGB, the base learners were split equally between 50 shared and 50 task-specific. All experiments were repeated five times, and the mean and standard deviation across all runs were reported. Specifically, the mean was first calculated over all tasks and then averaged across all repetitions.

4.2 Datasets

Three publicly available micro-level economic housing datasets were used in the experiments. A summary of the benchmark datasets, including the number of instances, features, and tasks derived from specific attributes, is provided in Table 2.

Each dataset provides housing information along with a target variable used to predict home prices. The *Ames* dataset includes features describing residential properties in Ames, Iowa, USA, with the target variable being the `SalePrice` of the houses. The *Boston* dataset contains data on housing values in the suburbs of Boston, Massachusetts, USA. Its target variable is `MEDV` (the median value of owner-occupied homes (in \$1,000s)). The `CHAS` feature, indicating proximity to the Charles River, was excluded due to its low variability and limited interpretability. The *California* housing dataset describes properties across various districts in California, USA, including summary statistics for each district, with the target variable being the `median_house_value`.

The selection of *Ames*, *Boston*, and *California* is deliberate and represents three distinct economic and housing environments, allowing us to evaluate the multi-task framework across heterogeneous market structures. *Ames* represents a detailed micro-level market with rich structural attributes and substantial within-city variation in construction quality and lot characteristics. *Boston*, however, captures an urban

Table 2 Descriptions of benchmarked housing datasets

Dataset	# Instances	# Features	# Tasks
Ames (Cock, 2011)	2,930	81	10
Boston (Harrison & Rubinfeld, 1978)	506	13	9
California (Kelley Pace & Barry, 1997)	20,640	9	5

environment where socioeconomic factors, environmental amenities, and accessibility play dominant roles in housing valuation. The *California* dataset offers a broader regional perspective, with large cross-district differences in demographics, income, and proximity to coastal areas. Hence, these datasets provide complementary settings that vary in granularity, dimensionality, and economic drivers.

Tables 3–5 summarize the descriptive statistics for the *Ames*, *California*, and *Boston* datasets, respectively. Table 3 reports descriptive statistics for the fifteen numerical features in the *Ames* dataset with the highest variability, as measured by the interquartile range (IQR). These variables capture key aspects of residential structures (such as lot size, basement area, living area, and construction year) and exhibit substantial dispersion across observations. For example, *Lot Area* ranges from 1,300 to over 215,000 square feet, illustrating the wide heterogeneity in parcel sizes within the Ames market. Similarly, measures such as *Gr Liv Area*, *1st Flr SF*, and *Total Bsmt SF* show large spreads that reflect diverse housing designs and construction standards.

Table 4 summarizes the distribution of the main numerical attributes in the *California* dataset. Unlike *Ames*, which relies on detailed structural characteristics, the *California* dataset captures aggregated district-level information such as total rooms, population, number of households, and median income. The wide ranges observed in variables such as *total_rooms* and *population* reflect the strong demographic and spatial disparities across regions. The geographic coordinates (*longitude* and *latitude*) also show substantial spread.

Table 5 presents descriptive statistics for the numerical features of the *Boston* housing dataset. Compared with *Ames* and *California*, this dataset is more compact but displays strong contrasts across socioeconomic and structural indicators. Variables such as *CRIM* (crime rate), *LSTAT* (lower status of the population), and *PTRATIO* (pupil-teacher ratio by town) exhibit substantial dispersion and long right

Table 3 Descriptive statistics of the top-variability numerical housing features in the Ames dataset. The table reports the fifteen features with the highest interquartile range (IQR), summarizing their central tendency and dispersion

Variable	Count	Mean	Std	Min	25%	50%	75%	Max
Lot Area	2930	10147.92	7880.02	1300.0	7440.25	9436.50	11555.25	215245.0
BsmtFin SF 1	2929	442.63	455.59	0.0	0.00	370.00	734.00	5644.0
2nd Flr SF	2930	335.46	428.40	0.0	0.00	0.00	703.75	2065.0
Gr Liv Area	2930	1499.69	505.51	334.0	1126.00	1442.00	1742.75	5642.0
Bsmt Unf SF	2929	559.26	439.49	0.0	219.00	466.00	802.00	2336.0
Total Bsmt SF	2929	1051.61	440.62	0.0	793.00	990.00	1302.00	6110.0
1st Flr SF	2930	1159.56	391.89	334.0	876.25	1084.00	1384.00	5095.0
Garage Area	2929	472.82	215.05	0.0	320.00	480.00	576.00	1488.0
Wood Deck SF	2930	93.75	126.36	0.0	0.00	0.00	168.00	1424.0
Mas Vnr Area	2907	101.90	179.11	0.0	0.00	0.00	164.00	1600.0
Open Porch SF	2930	47.53	67.48	0.0	0.00	27.00	70.00	742.0
MS SubClass	2930	57.39	42.64	20.0	20.00	50.00	70.00	190.0
Year Built	2930	1971.36	30.25	1872.0	1954.00	1973.00	2001.00	2010.0
Garage Yr Blt	2771	1978.13	25.53	1895.0	1960.00	1979.00	2002.00	2207.0
Year Remod/Add	2930	1984.27	20.86	1950.0	1965.00	1993.00	2004.00	2010.0

Table 4 Descriptive statistics of the numerical housing features in the California dataset. The table summarizes central tendency and dispersion across the full set of structural attributes used in the analysis

Variable	Count	Mean	Std	Min	25%	50%	75%	Max
total_rooms	20640	2635.76	2181.62	2.0000	1447.75	2127.00	3148.00	39320.00
population	20640	1425.48	1132.46	3.0000	787.00	1166.00	1725.00	35682.00
total_bedrooms	20433	537.87	421.39	1.0000	296.00	435.00	647.00	6445.00
households	20640	499.54	382.33	1.0000	280.00	409.00	605.00	6082.00
housing_median_age	20640	28.64	12.59	1.0000	18.00	29.00	37.00	52.00
longitude	20640	-119.57	2.00	-124.35	-121.80	-118.49	-118.01	-114.31
latitude	20640	35.63	2.14	32.54	33.93	34.26	37.71	41.95
median_income	20640	3.87	1.90	0.4999	2.5634	3.5348	4.74325	15.0001

Table 5 Descriptive statistics of the numerical housing features in the Boston dataset. The table summarizes central tendency and dispersion across the full set of structural attributes used in the analysis

Variable	Count	Mean	Std	Min	25%	50%	75%	Max
TAX	506	408.24	168.54	187.00000	279.000000	330.000000	666.000000	711.0000
AGE	486	68.52	28.00	2.90000	45.175000	76.800000	93.975000	100.0000
B	506	356.67	91.29	0.32000	375.377500	391.440000	396.225000	396.9000
RAD	506	9.55	8.71	1.00000	4.000000	5.000000	24.000000	24.0000
INDUS	486	11.08	6.84	0.46000	5.190000	9.690000	18.100000	27.7400
ZN	486	11.21	23.39	0.00000	0.000000	0.000000	12.500000	100.0000
LSTAT	486	12.72	7.16	1.73000	7.125000	11.430000	16.955000	37.9700
CRIM	486	3.61	8.72	0.00632	0.081900	0.253715	3.560263	88.9762
DIS	506	3.80	2.11	1.12960	2.100175	3.207450	5.188425	12.1265
PTRATIO	506	18.46	2.16	12.60000	17.400000	19.050000	20.200000	22.0000

tails. Conversely, accessibility indicators such as RAD (index of accessibility to radial highways) and environmental measures like NOX (nitric oxides concentration (parts per 10 million)) show concentrated but meaningful variation that is highly relevant for housing valuation.

Building on the descriptive statistics in Tables 3–5, it is also important to examine how these features differ in scale and completeness. Such differences directly affect model stability, preprocessing requirements, and the behavior of shared versus task-specific components in a multi-task framework. To this end, Fig. 1 provides a comparative visualization of feature dispersion and missingness for each dataset. It displays the distributions of feature scales and the proportion of missing values for all numerical variables. For improved interpretability, IQRs are log-scaled and normalized, enabling a clearer comparison across features that vary substantially in their economic units. In each horizontal bar, the blue segment represents the scaled IQR, while the red segment indicates the proportion of missing values. Notably, the *Ames* dataset exhibits wide disparities due to its mix of structural, quality, and categorical attributes. These insights support preprocessing steps such as scaling, encoding, and missing values handling to ensure stable learning across diverse feature spaces.

Regarding task construction, a task index was created by encoding categorical features into numerical indices representing different tasks. Specifically, the features used were Overall Qual (Overall quality) in *Ames*, ocean_proximity in *California*, and RAD in *Boston*. This was implemented by converting the chosen fea-

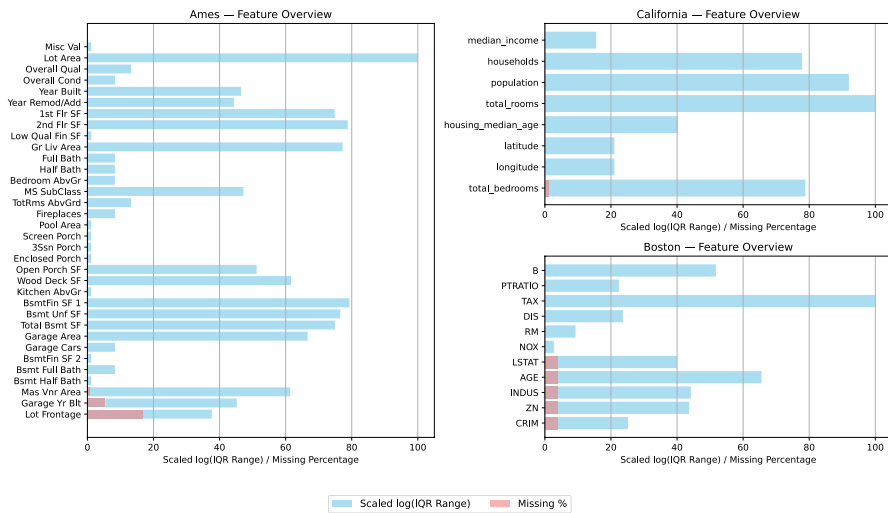


Fig. 1 Feature scale and missing data in benchmarked datasets (log-scaled)

ture to a categorical type and then encoding it to numerical indices. Figure 2 presents a comparative visualization of task-level structures across the three datasets: *Ames* (left subplot), *California* (center subplot), and *Boston* (right subplot). The horizontal axis represents the number of observations per task (*task sample size*), while the vertical axis shows the average target value within each task. Each bubble represents a single-task (Overall Qual in *Ames*, ocean_proximity in *California*, and RAD in *Boston*) with the bubble size proportional to the within-task variance of the target variable. Colors differentiate datasets, and the overall pattern highlights heterogeneity in both sample sizes and target distributions across tasks.

To evaluate the degree of relatedness among tasks and to motivate the use of a shared representation, a cross-task correlation analysis was conducted for each dataset. Specifically, for every categorical subgroup defining a task (e.g., Overall quality in *Ames*), the mean of each numerical feature was computed within the subgroup. This aggregation produced a task-level feature matrix, where each row corresponds

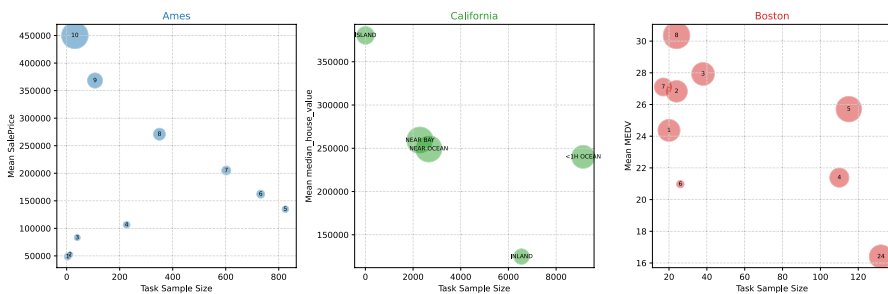


Fig. 2 Task-level summary bubble chart for Ames, California, and Boston datasets

to a task and each column represents the average value of a specific economic or structural attribute (e.g., as lot area, number of rooms, or construction year in *Ames*). Using this matrix, pairwise correlations between all aggregated features were calculated based on the Pearson correlation coefficient,

$$\text{corr}_{A,B} = \frac{\text{cov}(A, B)}{\sigma_A \sigma_B},$$

where, $\text{cov}(A, B)$ is the covariance between A and B , and σ_A and σ_B are the standard deviations of A and B , respectively. This quantity measures the linear association between A and B standardized by their variances. Pearson's coefficient was selected due to its interpretability and sensitivity to linear co-movements across tasks. This procedure enables the analysis of whether different subgroups in corresponding tasks (e.g., neighborhoods or regions) exhibit consistent co-variation patterns in their mean structural characteristics. High positive correlations indicate that changes in one feature (such as property size or overall quality) tend to correspond with proportional changes in others *across tasks*, suggesting the presence of latent shared economic mechanisms. Conversely, weak or negative correlations reveal more unique subgroup behavior. Across all three datasets, the correlation structures (the leftmost subplots in Figs. 3, 4, and 5) reveal clear and economically intuitive relationships among the numerical features, highlighting both shared determinants of housing values and meaningful differences across markets. The resulting correlation matrices provide empirical support for a multi-task learning framework: the tasks are neither independent nor fully homogeneous, but instead exhibit partially correlated structures.

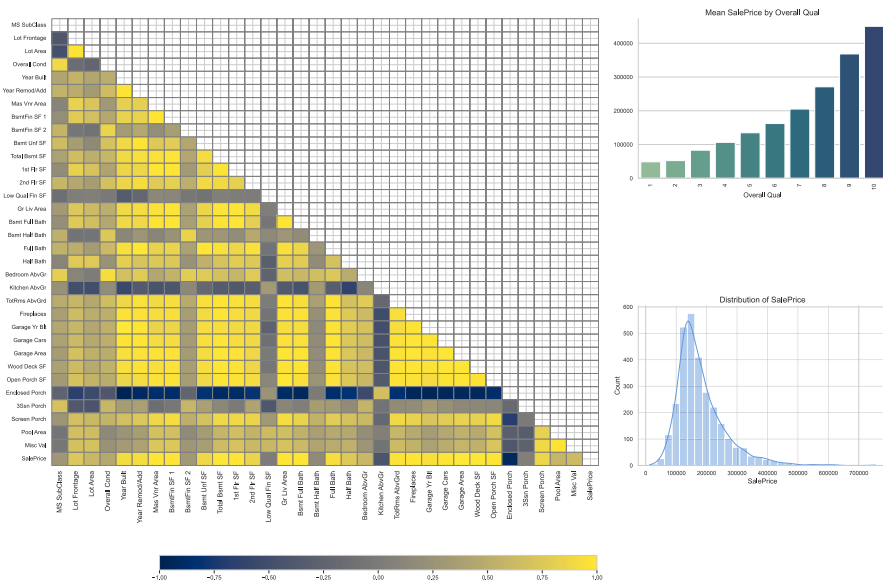


Fig. 3 Ames dataset: overall quality-level correlation structure, mean housing prices by overall quality, and overall price distribution

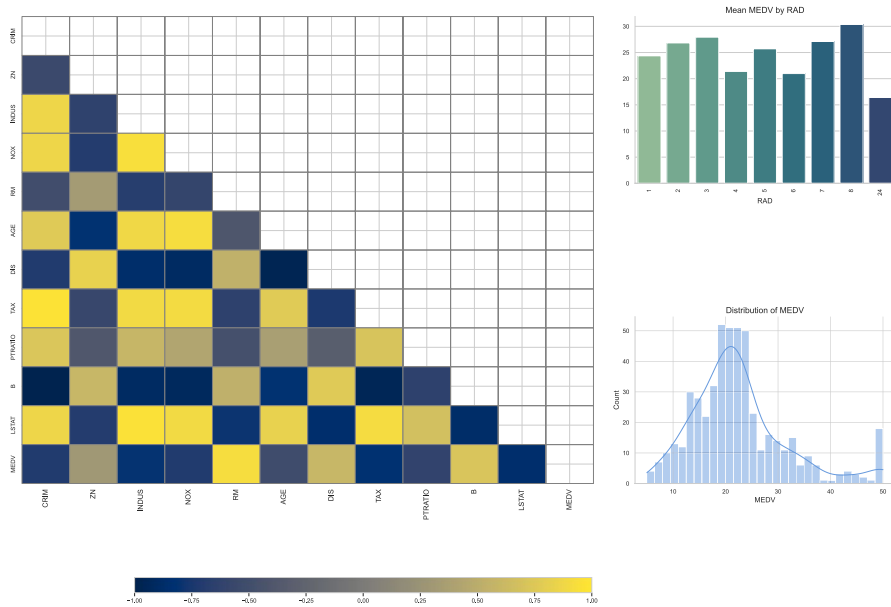


Fig. 4 Boston dataset: inter-task correlations across accessibility zones, mean home prices by highway accessibility, and target value distribution

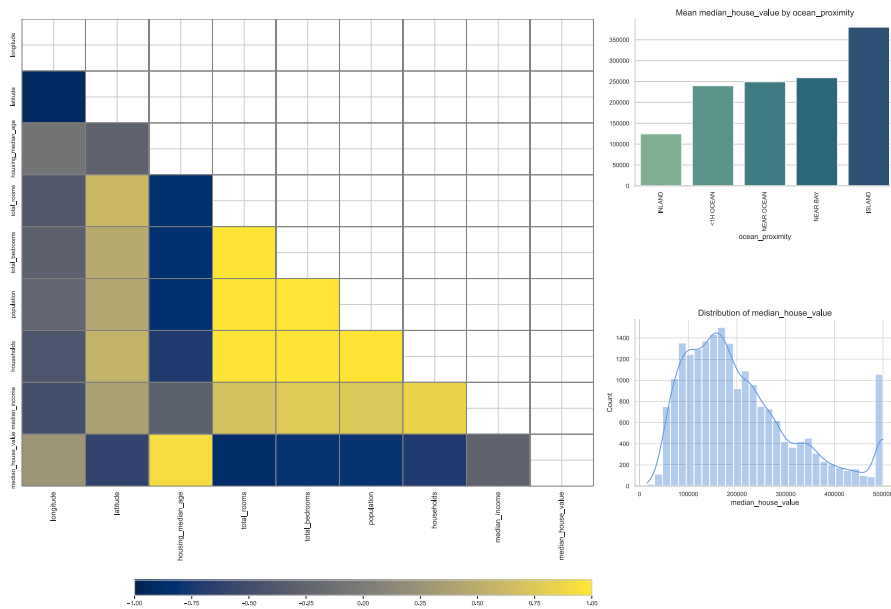


Fig. 5 California dataset: regional correlation patterns, mean housing values by ocean proximity, and distribution of median house values

The second (top right) subplot in each figure (Figs. 3–5) illustrates the mean value of the target variable within each task, providing a direct view of economic heterogeneity across subgroups. For instance, in the *Ames* dataset, average housing prices vary systematically according to overall quality, reflecting disparities in property characteristics and construction quality. Similarly, in the *California* dataset, the mean median house values differ across ocean proximity categories. These variations justify the inclusion of task-specific components in the model, as they demonstrate that while subgroups share common determinants of the outcome, each task retains unique influences that cannot be fully explained by global patterns alone.

The third (bottom right) subplot in Figs. 3–5 displays the global distribution of the target variable for each dataset, providing a macroscopic view of the economic variability captured by each dataset. The distributions display a clear skewed pattern. This asymmetry reflects the unequal distribution of economic value within regional housing markets, where a small number of high-value units coexist with a majority of moderately priced dwellings. These distributional patterns are consistent across the studied datasets and show that the target variable varies both within and across tasks.

4.3 Evaluation Metrics

The performance of the studied models was evaluated using standard regression metrics computed at the task-level. Specifically, we report the Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R^2), all calculated by comparing the predicted outputs of the studied models with the corresponding ground-truth values for each task. MSE and RMSE quantify the average squared prediction error and its square root, respectively, while MAE measures the average absolute deviation between predictions and observed values; for these error-based metrics, lower values indicate better performance. The R^2 metric captures the proportion of variance in the target variable explained by the model, where higher values (closer to 1) reflect better goodness of fit. For the overall evaluation, each metric was averaged across all tasks to ensure a consistent comparison of performance. All reported results were obtained on a held-out test set to assess model generalization and robustness.

4.4 Results and Discussion

For each dataset, both the mean and standard deviation across repetitions are presented in separate tables. For the Tables 6–8, the mean performance across all tasks is reported first, followed by the mean across all repetitions to provide a comprehensive view of model stability. The best results per-metric are highlighted using **boldface**. For the *Ames* dataset as reported in Table 6, MTGB achieves superior performance

Table 6 Model performance comparison for *Ames*, with best results per row in bold

Metric	MTGB	GB	CatBoost	LightGBM	XGB
MAE	0.028499	0.033545	0.034117	0.033928	0.033520
MSE	0.001684	0.002387	0.002510	0.002518	0.002416
R^2	0.847770	0.780771	0.769137	0.768827	0.778223
RMSE	0.040670	0.046994	0.047933	0.048331	0.047290

Table 7 Model performance comparison for *Boston*, with best results per row in bold

Metric	MTGB	GB	CatBoost	LightGBM	XGB
MAE	0.058469	0.062912	0.066273	0.095436	0.073352
MSE	0.008916	0.009714	0.011806	0.019958	0.015286
R^2	0.812490	0.789455	0.747726	0.565441	0.648609
RMSE	0.091778	0.096242	0.107109	0.140275	0.122227

Table 8 Model performance comparison for *California*, with best results per row in bold

Metric	MTGB	GB	CatBoost	LightGBM	XGB
MAE	0.101508	0.094989	0.096189	0.096296	0.094294
MSE	0.021209	0.022992	0.022831	0.024070	0.022063
R^2	0.625871	0.593984	0.596407	0.574267	0.609995
RMSE	0.144633	0.149167	0.148944	0.152454	0.146534

across all evaluation metrics, demonstrating the lowest MAE, MSE, and RMSE, as well as the highest R^2 . The improvement over single-task boosting methods confirms that leveraging multi-task representations is particularly advantageous in settings with structured and heterogeneous features such as housing attributes and construction quality.

Regarding the *Boston* housing dataset reported in Table 7, MTGB again outperforms all other models, achieving the best results for MAE, MSE, RMSE, and R^2 . This consistent advantage across multiple metrics underscores the ability of the multi-task ensemble approach to generalize better than independent single-task learners in compact, heterogeneous datasets.

For the *California* dataset (Table 8), results showed a more competitive scenario. While XGB slightly outperformed MTGB in terms of MAE, MSE, and RMSE, MTGB achieves the highest R^2 score (≈ 0.6259), indicating better overall explanatory power. This suggests that MTGB captures inter-task dependencies between regions with varying proximity to the ocean more effectively, even when pointwise errors are marginally higher. These results highlight the ability of the MTL application to exploit shared structures across tasks, leading to significantly enhanced predictive accuracy and robustness. In such settings, single-task models like XGB (XGBoost) tend to optimize performance around the densest regions of the target distribution, yielding marginal gains in MAE and RMSE by fitting the central mass of observations more tightly. By contrast, shared-plus-task-specific structure of MTGB enables partial pooling across regions, allowing the model to capture common valuation mechanisms while preserving task-level deviations associated with coastal and near-coastal markets.

To analyze model performance more in depth, per-task results for each model were examined using the box plots presented in Fig. 6. The plots show the distribution of per-task scores for each model (X-axis), with each box summarizing the variation of scores across tasks within a dataset. Two subplots were generated side by side: one for RMSE (left) and one for MAE (right). The per-task performance metrics were averaged across all repetitions to account for variability in training and evaluation. As shown in the plots, lower RMSE and MAE values were generally observed for MTGB across most tasks, in contrast to the single-task baselines (GB, CatBoost, LightGBM, XGB), which were trained independently for each task. Furthermore,

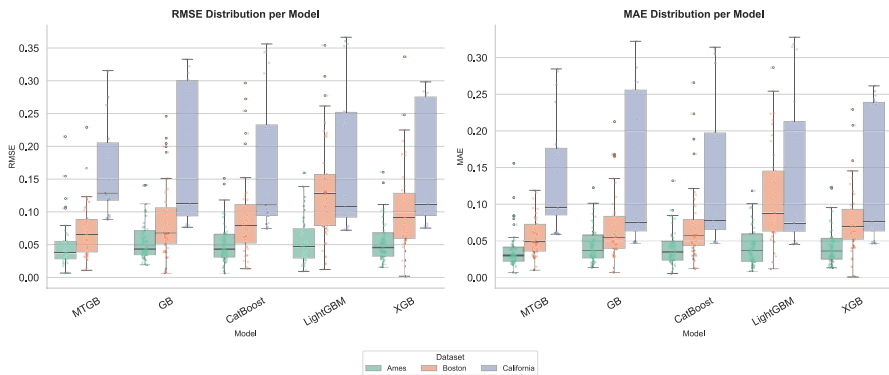


Fig. 6 Distribution of per-task RMSE and MAE across models

even for tasks with smaller sample sizes, competitive performance was maintained by MTGB.

To complement the pointwise error metrics (e.g., RMSE and MAE), we further examine how well each model captures the full empirical distribution of housing prices. This analysis is particularly important for economic datasets, where distributional shape reflects underlying market mechanisms, local heterogeneity, and non-linear structural patterns. For each dataset and model, we compare the estimated distribution of predicted values with the real distribution using kernel density estimates (KDE). We report the results for one batch of experiments on the *California* dataset as a representative example. Figure 7 provides a visual comparison that highlights the extent to which the models reproduce key distributional features, such as modality and dispersion. As evident in Fig. 7, MTGB predicted values demonstrate a closer alignment with the real distributions compared to single-task baselines. Among the single-task baselines, CatBoost shows the closest match, while GB and XGB exhibit larger deviations.

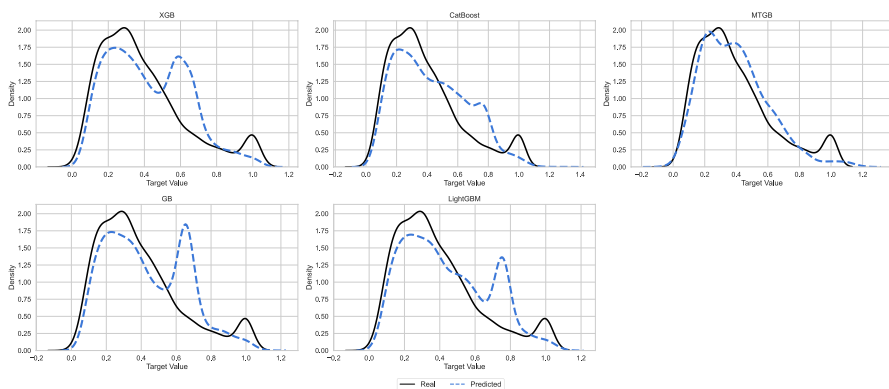


Fig. 7 Comparison of true and model-predicted housing price distributions for the California dataset across studied models (shown in separate subplots)

To formalize the visual comparison, we quantify distributional similarity using two well-established statistical metrics: (i) the Kolmogorov-Smirnov (KS) statistic (Massey, 1951), which measures the maximum distance between the empirical cumulative distributions of real and predicted values, and (ii) the Wasserstein distance (Villani, 2009), which evaluates the optimal transport cost required to align both distributions. In one representative batch of experiments, the corresponding values for each model and dataset are reported in Tables 9, 10, and 11, where the KS p-value indicates the statistical significance of the observed KS statistic. Lower values of these metrics indicate a better match between the distributions of the real and predicted values. The results reported in Tables 9–11 confirm that MTGB achieves the smallest distributional discrepancy across all datasets (highlighted in **boldface**).

Table 9 Distributional similarity metrics comparing predicted and true target distributions across models for the Ames dataset. Lower values for KS statistic and Wasserstein distance indicate higher similarity; higher KS p-values indicate weaker evidence against the null hypothesis of equal distributions. Best results per column are highlighted in bold

Model	KS Statistic	KS p-value	Wasserstein Distance
CatBoost	0.150171	0.000004	0.023696
GB	0.121160	0.000362	0.021790
MTGB	0.105802	0.002814	0.014788
XGB	0.126280	0.000172	0.022334
LightGBM	0.143345	0.000011	0.022899

Table 10 Distributional similarity metrics comparing predicted and true target distributions across models for the Boston dataset. Lower values for KS statistic and Wasserstein distance indicate higher similarity; higher KS p-values indicate weaker evidence against the null hypothesis of equal distributions. Best results per column are highlighted in bold

Model	KS Statistic	KS p-value	Wasserstein Distance
CatBoost	0.107843	0.595756	0.031054
GB	0.098039	0.713647	0.025461
MTGB	0.088235	0.824655	0.024647
XGB	0.098039	0.713647	0.028420
LightGBM	0.147059	0.220671	0.052762

Table 11 Distributional similarity metrics comparing predicted and true target distributions across models for the California dataset. Lower values for KS statistic and Wasserstein distance indicate higher similarity; higher KS p-values indicate weaker evidence against the null hypothesis of equal distributions. Best results per column are highlighted in bold

Model	KS Statistic	KS p-value	Wasserstein Distance
CatBoost	0.126453	3.63×10^{-29}	0.056449
GB	0.143411	2.03×10^{-37}	0.061041
MTGB	0.061531	3.23×10^{-7}	0.030569
XGB	0.136628	5.42×10^{-34}	0.056507
LightGBM	0.125000	1.66×10^{-28}	0.064008

5 Economic Interpretation and Insights

Beyond predictive performance, the task-level evaluation of MTGB reveals meaningful economic structure across the datasets. The shared learning component captures broad valuation mechanisms that operate similarly across submarkets (such as overall construction quality or general demand conditions) while the task-specific components model local adjustments that account for heterogeneity across regions or categories. The variation in task-level performance (e.g., differences in RMSE across tasks) reflects the extent of such heterogeneity: tasks with lower errors correspond to submarkets (tasks) whose price formation is well explained by the shared component, whereas tasks with higher residuals represent regions characterized by stronger local idiosyncrasies.

Figure 8 illustrates this behavior for the studied datasets (depicted across multiple subplots), where each point corresponds to a distinct categorical subgroup (task). The smoother trajectory and systematically lower RMSE values obtained by MTGB across most tasks, relative to single-task baselines such as XGB or CatBoost, indicate that knowledge learned in one subgroup is successfully transferred to others. This pattern aligns with the interpretation of MTGB model, combining a global valuation structure with task-level corrections that reflect regional heterogeneity. In economic terms, these findings support the interpretation of MTL framework as a hybrid between a pooled and a fully disaggregated model. It learns a global structure while allowing each subgroup to deviate through task-specific learners. The decomposition thus provides an interpretable bridge between ensemble MTL and traditional econometric reasoning.

6 Conclusion

This study presented the first systematic application of Multi-Task Gradient Boosting (MTGB) to economic data. The empirical results demonstrated that the multi-task framework can effectively exploit shared representations across economic segments,

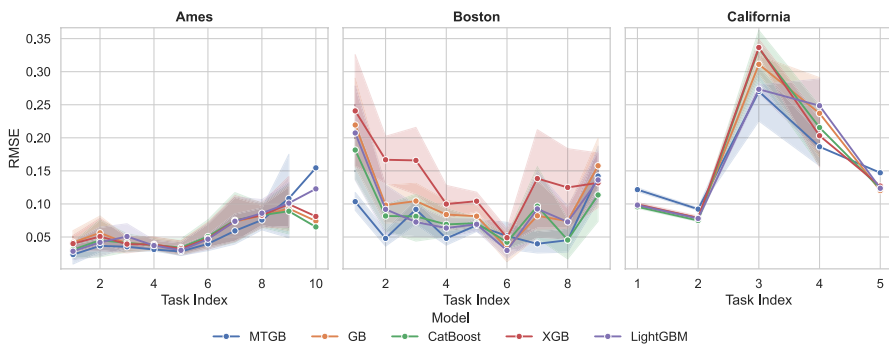


Fig. 8 Distribution of per-task RMSE across models (indicated by colors) and datasets (shown in separate subplots)

leading to improved predictive accuracy and interpretability compared to single-task ensemble models such as XGB, LightGBM, and CatBoost.

Across benchmark housing datasets, MTGB consistently achieved higher stability and stronger generalization performance, particularly in scenarios with interrelated but heterogeneous subgroups. These findings confirm that learning shared representations is beneficial for modeling economic processes that exhibit both common and task-specific components. The results further indicate that the MTGB framework can uncover latent dependencies between regional or categorical units, such as housing markets with structural and locational attributes. The *California* housing results highlight that performance differences across models are partly driven by the underlying distributional properties of the data. In markets characterized by heterogeneity and skewed distributions, pointwise error metrics such as MAE and RMSE may favor models that fit the central mass of observations, while variance-based measures such as R^2 score better reflect a model's ability to capture differences across subgroups. The higher R^2 score achieved by MTGB in this setting indicates that the shared multi-task structure is particularly effective at explaining between-task variation arising from structural and locational heterogeneity, even when marginal prediction errors remain comparable.

The empirical analysis across the studied datasets further demonstrates that the advantages of MTGB are not tied to a single dataset or market structure. These datasets represent three different economic environments—micro-level structural variation, urban heterogeneity, and broad regional demographic and geographic diversity. Consistent performance gains across all three settings show that the shared representation learned by MTGB captures latent economic mechanisms that generalize beyond a specific dataset. This confirms that the observed improvements are not a byproduct of a particular housing market, but reflect the ability of the multi-task model to exploit partially correlated structures across diverse subgroups, validating the suitability of multi-task learning (MTL) for non-homogeneous economic data.

Empirically, MTGB shows that ensemble-based MTL can serve as an alternative to both traditional econometric models and neural approaches, combining predictive performance with transparency. Future research could extend this framework to firm-level or sectoral data, incorporating temporal dependencies, causal inference mechanisms, or policy simulations within the multi-task paradigm.

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Availability of Data and Materials All data analyzed in this study are publicly available and cited in the paper.

Declarations

Conflicts of Interest The authors declare that they have no conflict of interest.

Ethics Approval Not applicable.

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